Palm Target Positioning

1. Course Content

This course implements obtaining color images and using the mediapipe framework to detect palms and output palm coordinates.

2. Program Startup

For Raspberry Pi 5 users, if the ROS container is not started, you need to start the container first,

```
bringup_m3
```

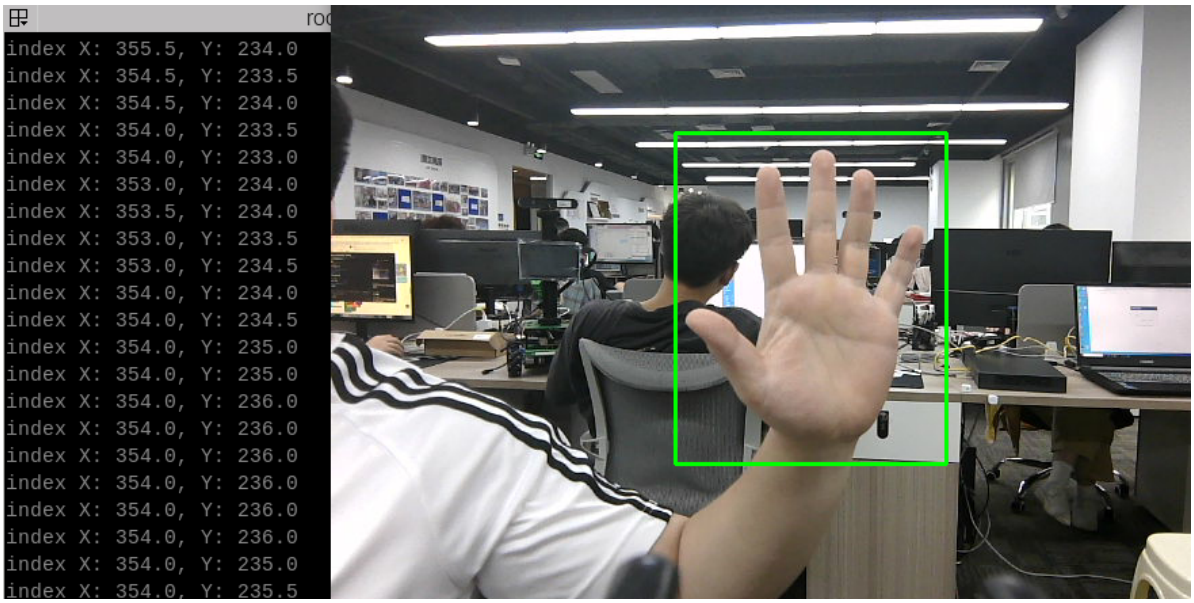
Open a terminal inside the container,

```
exec_m3
```

- Run program in terminal

```
ros2 launch m3_bringup mediapipe_demo.launch.py demo:=FindHand
```

After the program starts, as shown in the figure below, when a palm is detected, it will be framed in green in the image, and the palm center coordinates will be output in the terminal.



3. Core Code Analysis

Program code path:

```
~/yahboomM3_ws/code_ws/src/m3_common_demo/m3_common_demo/mediapipe/10_FindHand.py
```

Import the required library files,

```
import rclpy
from rclpy.node import Node
from M3Pro_demo.media_library import *
import cv2 as cv
import numpy as np
import time
import os
import threading
from cv_bridge import CvBridge
from sensor_msgs.msg import Image
from arm_msgs.msg import ArmJoints
import cv2
```

Initialize data and define publishers and subscribers,

```
def __init__(self,name, mode=False, maxHands=2, detectorCon=0.5, trackCon=0.5):
    super().__init__(name)
    #Call the media_library library to create an object of the HandDetector
class
    self.hand_detector = HandDetector()
    #create a publisher
    self.rgb_bridge = CvBridge()
    #Define the subscriber for color image topic
    self.sub_rgb =
self.create_subscription(Image,"/camera/color/image_raw",self.get_RGBImageCallBa
ck,1)
```

Color image callback function,

```
def get_RGBImageCallBack(self,msg):
    #Use CvBridge to convert color image message data to image data
    rgb_image = self.rgb_bridge.imgmsg_to_cv2(msg, "bgr8")
    #Pass the obtained image to the process function for palm detection
    frame = self.process(rgb_image)
    key = cv2.waitKey(1)
    cv.imshow('dist', frame)
```

process function,

```
def process(self, frame):
    #Call the object's method to perform palm detection, returning the detected
image as well as lmList list and bbox list
    frame, lmList, bbox = self.hand_detector.findHands(frame)
    self.hand_detector.draw = True
    #If the lmList list is not empty, it means a palm was detected
    if len(lmList) != 0:
        hand = self.hand_detector.fingersUp(lmList)
    #Get the xy coordinates of the palm, bbox stores the xy coordinates of the
top-left and bottom-right corners of the palm
    indexX = (bbox[0] + bbox[2]) / 2
    indexY = (bbox[1] + bbox[3]) / 2
    print("index X: %.1f, Y: %.1f" % (indexX, indexY))
    return frame
```

Regarding the definition of the Medipipe recognition class, it can be found in the media_library.py library. The location of this library is in the m3_common_demo package with the following path:

```
~/yahboomM3_ws/code_ws/src/m3_common_demo/m3_common_demo/mediapipe/media_library
.py
```

When needed, you can pass parameters into it. For example, we define the following functions in the HandDetector class:

- findHands: Find palms
- fingersUp: Fingers extended upward
- ThumbTOforefinger: Detect the angle between thumb and index finger
- get_gesture: Detect gestures